3 History of the Study of Human Biology

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INTRODUCTION

The field of human biology is broad based but with its principal origins in studies of variation in living populations within physical or biological anthropology. Today, human biology incorporates a majority of scientists trained in anthropology, but also counts among its members scientists trained in other specialties of the human sciences. During the late nineteenth and early twentieth centuries, our stem science – physical anthropology – was oriented toward skeletal studies, gross anatomy, and human variation as represented by race. Skeletal studies included those of both living and prehistoric populations. Anatomically oriented studies of the skeleton and of the living were focused on structure and origins, but with less interest in function and evolutionary causality. In his *Manual of Physical Anthropology*, Juan Comas (1960) presented an excellent overview of the history of physical anthropology from its earliest origins, including the origins of and connections with natural history, racial classification, craniometry, prehistory and paleoanthropology, and evolution. In later chapters, he reviewed the histories of growth studies, somatology, constitutional typology, craniology, osteology, and racial classification.

The beginning of an integrated human biology in the United States dates back to the late 1920s when Darwin's ideas of population variation, adaptation, selection, and evolution began to be reconceptualized by a number of leading scientists. At that time, evolutionary theory, and concepts from behavioral sciences, demography, genetics, and child growth began to be consolidated into a science of human biology. In later decades, body composition, physiology, nutrition, and ecology were added to the mix leading to an even greater understanding of the variations in, the evolution of, and the characteristics of humans in all of their complex biobehavioral states.

A history of anything is a reflection of *traditions*, *connections*, and *innovations*. These can be found as well in a history of human biology. In the narrative that follows, the development of the science of human biology will be traced through the contributions of key authors, their ideas, and the role of key institutions in its history.

RACE AND TYPOLOGY

The tangled roots of interests in human biology and human variation lie deep in antiquity. In the early modern era, Europeans became acutely aware of human population variation during the Age of Exploration and the opening of new regions of the world in the fifteenth, sixteenth, and seventeenth centuries (Eiseley, 1958). Newly discovered populations from the Americas, Africa, the Pacific, and Asia, were classified, along with Europeans, into broad racial groups according to physical characteristics. Such racially based physical characteristics were erroneously thought to be tightly bound to mental, emotional, intellectual, and cultural attributes, as well. Some races were identified as clearly inferior to others on a typological scale – an extension of *The Scale of Being* – from primitive to advanced. Ideas of fixed, unchanging racial categories were tightly bound to Judeo-Christian religious beliefs associated with, among other things, the creation or origin of humans. Some believed that humans originated from a single creation, *monogenism*, where the more primitive races had then degenerated from the original people living in the Garden of Eden. Another belief was in *polygenism*, where God had created several independent peoples, some who were elevated to civilization and others doomed at the outset to perpetual savagery. In both of these beliefs of human creation, Western, civilized populations were held, with a few exceptions, to be superior to non-Western races from Asia, Africa, the New World, and the Pacific.

Within Western societies, the two sexes were considered unequal – men were believed to be superior to women, by virtue of their very nature, and women were often viewed as incapable of rational thought (Stocking, 1987). Women were more emotional than men and tied more closely to nature through their role in reproduction

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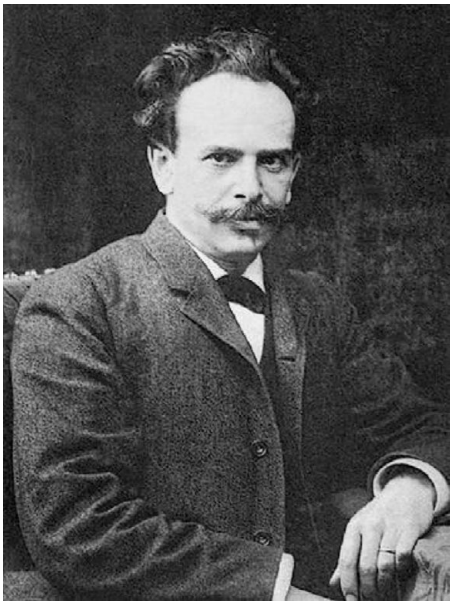
and childbirth. The upper socioeconomic classes were believed to be innately superior to the lower classes, because class differences were thought to have a hereditary basis. Hence, aristocratic males ("good breeding") were alleged to be at the pinnacle of God's creations. There *were* some who questioned the dominance of nature over nurture, however. For example, in the last chapter of the popularly known *Voyage of the Beagle* (Darwin, 1839), a thoughtful Charles Darwin noted, in the context of slavery: "... if the misery of the poor be caused not by the laws of nature, but by our institutions, great is our sin ..." (Darwin, 1839, p. 433). Questions about the contributions of nature (heredity) and nurture (environment) to human diversity persist and are debated up to the present.

Eighteenth-century racial classifications were established by Carolus Linnaeus (1707–1778), Georges Louis Leclerc, the Comte de Buffon (1707–1788), Georges Cuvier (1769–1832), and, of course, Johann Friedrich Blumenbach (1752–1840) (Gould, 1996; Molnar, 2002). Classifications ranged from Linnaeus's Asian, European, African, and American groups to Blumenbach's Caucasian, Mongolian, Ethiopian, American, and Malayan.

Twentieth-century classifications of human populations have been devised in numerous combinations and have been based on a variety of criteria. Earnest A. Hooton (1931), the Harvard professor who trained most of the modern generation of physical anthropologists before World War II, identified "composite races," in addition to the "primary races and subraces." The composite races were formed by admixture of primary races. Carleton Coon was one of Hooton's early students who had written a book classifying *The Races of Europe* (Coon, 1939). Several years later, Coon et al. (1950), in a mid-twentieth century approach, presented a six-fold geographical classification and then divided these major races into thirty subpopulations, whose characteristics were based on external physical and skull attributes. In that same year, Boyd (1950) identified six races according to their blood group genetics. About a decade later, Garn (1961), who was a coauthor of the Coon et al. (1950) book, refined these classifications and identified geographical, local, and micro-races, in a hierarchy of populations and subpopulations. These three works were different from previous efforts at classification in that they attempted to apply contemporary evolutionary, genetic, and ecological principles to the identification of racial (population) variation around the world. They were transitional in the sense that they applied modern theory to an outdated typological system in which boundaries between populations were fixed. There has been considerable controversy in scientific writings over the intervening years about the utility and social impact of racial classification and the validity of the concept itself. Works by Shipman (1994), Marks (1995), Brace (2005) and the first chapter by Mielke et al. (2006) provide histories of these controversies about race. Fundamentally, all classifications of this nature are *essentialist* in character, that is, they focus on fixed *types* that obscure the variation that contributes to human population diversity around the globe. It was the rejection of typological thinking (fixed species and races) that enabled Darwin to conceptualize ideas of variation, adaptation, and natural selection (Mayr, 1972), and it was this same transition in thinking about human populations that moved human biologists toward a truly scientific approach in this field.

FRANZ BOAS AND HIS CONTRIBUTIONS

Although the late nineteenth and early twentieth centuries were characterized by beliefs in fixed racial types, accompanied by beliefs in the superiority of some racial groups over others (racism or racialism), there was a contrary opinion in the form of Franz Boas (1858–1942). Boas, often referred to as the founder of American anthropology and the four-field approach, made remarkable contributions to dispelling the myth of fixed or pure races and the importance of the environment in structuring the character of human populations (Figure 3.1). These interests in human plasticity in the context of race were almost certainly based on his research on child and adolescent growth and development (Tanner, 1959, 1981). Hence, he can be identified also as one of the founders of human biology in the United States because of his contributions to: (1) debunking the idea of fixed races; (2) establishing a migration research design that



3.1. Franz Boas (1858–1942) in 1906. Photograph with permission from the American Philosophical Society.

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continues to be used up to the present; (3) incorporating the social and material environment as influencing human biology (plasticity); and (4) making numerous discoveries about the patterns of growth in children and adolescents.

The migrant study, designed to test the idea of races as static types, was initiated in 1908 with modest funding from the US Immigration Commission. The pilot study began in June 1908 with studies of height and cephalic index of Eastern European Jewish boys in New York City Schools (Stocking, 1974). The results of the pilot study and the more extensive study of Bohemians, Sicilians, Neapolitans, Polish, Hungarians, and Scots established differences between those born to parents before and those born to parents after they migrated to the United States (Tanner, 1959). Head form or cephalic index differed between the two groups, refuting the idea of fixed races and demonstrating the influence of the environment on human variation in physical characteristics. The completed study, published by Boas (1912), was accompanied by the publication several years later of the raw data of more than 18 000 subjects who were measured in the migration study (Boas, 1928). The reanalysis of these data stimulated a new controversy (Sparks and Jantz, 2002; Gravlee et al., 2003) over Boas's analyses and interpretations and whether he really demonstrated plasticity in these migrant populations.

Relethford (2004), in exploring this controversy, identified three ways that craniometric variation can change over time: (1) developmental plasticity through environmental change; (2) long-term changes through natural selection; and (3) within-group and among-groups variation by gene flow. Each of these has been shown to operate, but what Relethford (2004) noted was that the debate about Boas's study centered on the relative importance of these three causes of craniometric change. Based on the two major studies, the question that Relethford raised, "… is whether developmental plasticity has a significant effect on craniometric variation" (Relethford, 2004, p. 380). Relethford approached this question by inspecting the changes in craniometric variation among the groups Boas studied. Three of the seven European ethnic groups showed no statistical difference between US-born and European-born migrants (Hungarians, Polish, and Scots). The remaining four groups did show statistically significant differences, but these were relatively slight differences. Based on this, Relethford (2004) suggested that these data *do* demonstrate developmental plasticity, but this plasticity does not obscure the underlying genetic differences that separate the ethnic groups. In other words, both genetic characters and developmental plasticity contribute to the variation, but the genetic contribution to the total variation, in this case, is the stronger of the two. Relethford's (2004) conclusion was anticipated by Boas in a paper published years after the original, where Boas acknowledged the plasticity in the immigration study when he stated: "These changes do not obliterate the differences between genetic types [characters] but they show that the type as we see it contains elements that are not genetic but an expression of the influence of the environment" (Boas, 1936, p. 523).

Boas's growth studies date back to 1888 when he took an academic position at Clark University in Worcester, Massachusetts. His first important discovery was based on data gathered by the growth studies pioneer and Harvard professor, Henry Bowditch (Tanner, 1959). Boas discovered that the asymmetrical distributions in Bowditch's data on height during adolescence could be explained by the individual variations in growth rates, which he called "tempo of growth." This is the first indication of Boas's sensitivity to the importance of longitudinal growth data in uncovering subtle changes in growth during childhood and adolescence. In 1891, he initiated a longitudinal survey of Worcester school children that confirmed his sense of the value of longitudinal growth data. Between 1892 and 1941, Boas published numerous papers on growth in *Science*, *Human Biology*, and other journals in which he made other significant discoveries, each a reflection of his remarkable knowledge of statistics and his genius. Over the years, he: (1) produced the first National Growth Standards or norms; (2) introduced the concept of *physiological or developmental age* and observations that males were behind females as early as five years of age; (3) observed that working class and poor children from large sibships tend to be smaller on average than those from small sibships; (4) established relationships between "age of peak velocity" during adolescence and other measures of size during adolescence and adulthood; and (5) observed that children from the Horace Mann School of Columbia University had become larger between 1909 and 1935 (now known as the *secular trend in growth*) (Tanner, 1959, 1981).

PRE-WORLD WAR II AND HUMAN POPULATION BIOLOGY

In addition to Boas's contributions to a developing field of human biology through his growth studies, there are several other lines of continuity from pre-World War II to the present. In 1929, Raymond Pearl (1879–1940) founded the journal *Human Biology* and served as editor until his death in 1940 (Crawford, 2004). Pearl was an accomplished population biologist who was Professor of Biometry and Vital Statistics in the School of Hygiene and Public at Health at Johns Hopkins University and who had broad interests in genetics, fertility, evolution, nutrition, disease, duration of life, senescence, and physical anthropology (Figure 3.2). According to Kingsland (1984, p. 8), "Pearl considered himself to be first

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3.2. Raymond Pearl (1879–1940) sometime in the late 1920s or early 1930s. Photograph with permission from the American Philosophical Society.

and foremost a human biologist"; that is, he was totally committed to a human population biology that was quite akin to some of the holistic and integrated science practiced today.

The journal, *Human Biology*, published a variety of papers in, among other topics, genetics, growth, health, human population studies, mathematical and statistical modeling, and human evolution. Goldstein (1940), in a survey of the first decade of *Human Biology* (*HB*) and two decades (1920s and 1930s) of the *American Journal of Physical Anthropology* (*AJPA*), suggested that the journal *HB* published articles more in the realm of "biological anthropology" whereas the *AJPA* was more prone to publish in "anatomical anthropology." These patterns of topical publication imply that the earliest development of a professional human biology "identity" began about this time. As Goldstein (1940) reported, there were strong connections between Pearl and his journal and physical anthropology – connections with the journal that have continued to the present. Many early papers in *Human Biology* were published on topics related to physical anthropology (Lasker, 1989). Raymond Pearl and Franz Boas were acquainted at that time because Pearl was active in a number of professional societies in which Boas was a member, including the American Association of Physical Anthropologists (AAPA). In fact Pearl was well known among physical anthropologists of this time as evidenced by his election as the third president of the AAPA from 1934 to 1936. Moreover, both Pearl and Boas were members of the National Academy of Sciences, both were sophisticated biostatisticians, and Boas published several papers on growth in the journal *HB* while Pearl was editor in the 1930s. It is probably the case that Aleš Hrdlička, one of the founders and first president of the AAPA, and Pearl did not share the same scientific philosophy because of Hrdlička's strong dislike and avoidance of statistics and Pearl's commitment to their use (Lasker, 1989). This added an additional dimension to the differences between the *AJPA* and *HB*, since Hrdlička was editor of the former and Pearl was editor of the latter.

Perhaps the most extensive pre-war research in human biology was that of human growth conducted at several centers in the United States. Some of these longitudinal growth studies (sequential measurements over time of the same individuals) were the Fels Longitudinal Study (Yellow Springs Ohio), the Bolton-Brush Study at Western Reserve University (Cleveland, Ohio), the Berkeley Growth Study at the Institute of Human Development, University of California (Berkeley), the Child Research Council Study at the University of Colorado (Denver), and the Harvard School of Public Health Growth Study (Boston) (Roche, 1992, pp. 1–2). Each of these was founded from the 1920s to the early 1930s principally to determine the effects of the Great Depression and to assess the means to help impoverished children. As Tanner noted: these longitudinal studies were part of ". . . a powerful child welfare movement [that] arose in the 1920s and provided the soil for a crop of longitudinal studies, whose harvesting shaped the whole pattern of human auxology in the years 1935–1955" (Tanner, 1981, p. 299). Here he uses the term *auxology*, which refers to the study of human physical growth, which he (James Tanner) pioneered in the 1950s and 1960s in the United Kingdom (Figure 3.3). It is not clear to what extent Boas's work

3.3. James M. Tanner (1920–) at a conference in 1982. Photograph courtesy of Barbara Garn.

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stimulated these longitudinal growth studies, but principle investigators must have known the value of longitudinal series from Boas's work before the turn of the twentieth century and later papers in the 1930s. For example, when Frank Shuttleworth (1889–1958) analyzed the Harvard Growth Study data, he employed some of the same procedures that Boas employed in analyzing data of adolescent girls (Tanner, 1981, p. 310).

Boas's influence was also seen in students from Harvard and elsewhere who used his migration model in their own research during the 1930s and 1940s (Little and Leslie, 1993). For example, Harry Shapiro and Frederick Hulse (1939) studied Japanese migrants to Hawai'i, Marcus Goldstein (1943) measured Mexican-Americans, and Gabriel Lasker (1946) studied immigrant and American-born Chinese. The migration-research design was also used in a great deal of human biology research up to the end of the twentieth century.

THE WORLD WAR II YEARS

The enormity of World War II brought a halt to a great deal of academic research in the United States and elsewhere, but stimulated research that could be directly applied to the war effort. Some of this research carried over to the post-war years and not only led to additional academic research, but even contributed to changes in thinking about the causes of human variation. Much of the wartime research centered on the maintenance and health of US troops in stressful environments and ameliorating the stress of warfare and hunger in the civilian populations in Europe and Asia. During the war, US military personnel were stationed and fighting in Europe, East Asia, Southeast Asia, North Africa, the Pacific, and Alaska and the Aleutian Islands. Hence, they were expected to be able to work strenuously at high levels while at the same time being exposed to climatic variation that ranged from Arctic and temperate zone winter cold to tropical and temperate summer heat. In addition to stresses from dietary restriction and starvation and from climatic extremes, there was interest in the interaction between diet and climate, particularly in the context of providing enough food calories for troops fighting in cold or hot geographic zones (Mitchell and Edman, 1951). Two of the most prominent wartime studies will be surveyed briefly: the semi-starvation study conducted at the University of Minnesota and the desert physiology study conducted by the Rochester Desert Unit whose research was conducted at several places in the United States.

The Minnesota semi-starvation study was initiated by Ancel Keys (1904–2004), a stress physiologist best known for having developed K-rations during the early part of the war, and several colleagues including Josef Brožek (1913–2004), a psychologist who later distinguished himself in studies of body composition. The study began in 1944 with 36 conscientious objectors who volunteered for the full-year study that consisted of an equilibration period on a normal diet, 3 months of semi-starvation (1600 kcal/day), and a period of refeeding. In addition, the volunteers were expected to walk 22 miles per week to increase their energy expenditure. The study was intensive, with almost daily biochemical, physiological, psychological, and body composition measurements taken. The results of the study were published in two massive volumes that stand as state-of-the-art research even today, particularly since it probably would not be permitted to conduct such a high-risk project during present times (Keys et al., 1950).

The Rochester Desert Unit was charged with the research task of determining water and food requirements, sweat rates, energy balance, heat tolerance, work capacity, and the probability of survival while living under the hot-dry conditions of desert environments. Much of the desert research was conducted on military personnel on maneuvers in southern California. Other studies were conducted of men on life rafts without water, and some limited tests were conducted under hot-wet conditions in Florida to simulate troops in tropical forests. In the preface to the volume that reported this research, the authors identify this work as a part of the "... recently developing field of environmental physiology" (Adolph and Associates, 1947, p. vii). What was not known then was that this work was to stand as the first major work in climatic stress physiology, and one that would serve as the basis for heat-stress studies both in environmental physiology and human biology.

In addition to the pioneering research in nutritional and environmental physiology represented by these two wartime studies, a relatively new area of human variation was being developed that would be explored by anthropologists and human biologists interested in growth, environmental stress, exercise physiology, and nutrition – *body composition*. These early studies introduced hard-tissue anthropologists (bone and teeth) to the importance of soft tissue (muscle and adipose tissue) in human function, human growth, and in studies of human adaptation to the environment.

POST-WAR UNITED STATES AND UNITED KINGDOM

There were parallel transformations of physical anthropology in the years directly following World War II in both the United States and in the United Kingdom. In the United States, Sherwood L. Washburn (1911–2000) successfully promoted a scientific agenda of problem solving, application of evolutionary theory, and understanding of human variation rather than racial

3.4. Sherwood L. Washburn (1911–2000) at the 1952 Wenner-Gren Foundation International Symposium of Anthropology. Photograph courtesy of the Wenner-Gren Foundation for Anthropological Research.

typology and simply descriptive anthropometric survey (Washburn, 1951). Washburn, who had been trained at Harvard, among many others, under Earnest A. Hooton (1887–1954), rejected many of the traditional ideas of his mentor and began an active program to bring his new ideas to other anthropologists, especially younger ones (Figure 3.4). With support from a private anthropological foundation in 1946, he organized the Viking Fund Summer Seminars in Physical Anthropology. These Summer Seminars were dynamic "state-of-the-art" meetings in which many new ideas in physical anthropology were explored for the first time. Washburn, despite having received his PhD degree only six years earlier (1940), already was a dynamic force in the field of physical anthropology. He was Secretary-Treasurer of the AAPA (1943–1946), had been teaching anatomy at Columbia University since 1939, and persuaded Paul Fejos (1897–1963), Director of the Viking Fund (later the Wenner-Gren Foundation), to sponsor these Summer Seminars held in New York City. In addition, Washburn enlisted a new Harvard PhD, Gabriel W. Lasker (1912–2002) to edit a new *Yearbook of Physical Anthropology* to report on the Summer Seminars and other news of the profession as well as to reprint important papers not easily found in American anthropology journals. The *Yearbook of Physical Anthropology* has continued to the present from the year of its first publication in 1946.

In 1950, the fifth year of the Summer Seminars, Washburn organized a major conference that served to bring together physical anthropologists and human geneticists with hopes to produce a fresh synthesis in human variation and evolution. The conference was the 15th Cold Spring Harbor Symposium on Quantitative Biology (Warren, 1951) and was initiated by the Cold Spring Harbor Institute Director, Milislav Demerec (1895–1966). Washburn's co-organizer of the Symposium, called "Origin and Evolution of Man," was the distinguished geneticist, Theodosius Dobzhansky (1900–1975), whom he had met at Columbia University. Although the conference was successful in bringing together geneticists, evolutionists, and anthropologists, there were some differences among the more traditional physical anthropologists, particularly the typological "constitutionalists" such as William Sheldon and Earnest Hooton on the one hand, and the more forward-looking physical anthropologists and geneticists on the other concerning topics of race and population variation. "Constitutional somatology" is an outmoded and typological area of study of body form and associated behavior that has a long history (see Comas, 1960, 319 ff.) and was promoted by Sheldon (Sheldon et al., 1940) and Hooton (1939) at the Cold Spring Harbor meeting. Many years later, the daughter of the Institute Director, Rada Dyson-Hudson, who had attended the Symposium as a young Swarthmore College student, remarked that after the afternoon sessions, the geneticists would gather to discuss the talks given that day, while the anthropologists would head for the nearest local tavern! Despite these differences, new ground had been broken, and Washburn's "scientific and evolutionary" physical anthropology was gaining momentum.

In the United Kingdom, the transformation also involved experimental scientific approaches to investigating human variation and the application of evolutionary theory as a basis for explaining human variation. The United Kingdom's transformation moved in two directions: first, toward an adaptive and ecological view of human biological function, and second toward an evolutionary view of human genetic population structure. The leaders in this movement were Joseph S. Weiner (1915–1982) and his colleague Nigel A. Barnicot (1914–1975) (Harrison, 1982). Weiner, who was trained in physiology, anatomy, and anthropology in South Africa, came to England in 1937. With post-war academic posts at Oxford with the distinguished anatomist Le Gros Clark, and later at the London School of Hygiene and Tropical Medicine, he contributed to the training of a whole generation of human biologists, promoted the transformation of physical anthropology, and was instrumental in organizing and managing human adaptability research during the International Biological Programme (IBP) from 1962 to 1974 (Harrison and Collins, 1982). Barnicot was trained in zoology and physiology, and was in the Anthropology Department at University College, London for most of his professional life. He worked on blood genetics, and skin and hair color in West African populations, on studies of nonhuman primates and, in later years, on the

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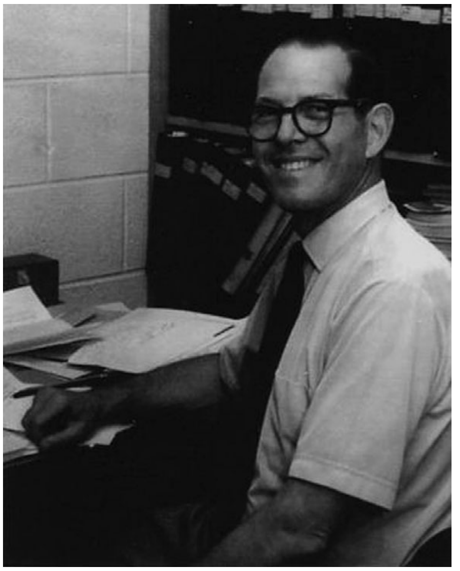
human biology of Tanzanian Hadza hunter-gatherers (Sunderland, 1975).

Joseph Weiner was also a strong supporter of biocultural approaches in human biology, in addition to his commitment to solid experimental studies in human physiology. In a 1958 paper on "... training in physical anthropology and human biology," he noted, "... it should be emphasized that in their [students'] research interests an important field of overlap exists between biological and cultural anthropologists" (Weiner, 1958, p. 47). Shortly before his death and nearly 25 years after the earlier statement, Weiner wrote:

> The simple fact that his unit of study is a defined community has made it imperative for the human biologist to take full account of the sociocultural properties of his community. Population structural analysis ... cannot be pursued without close attention to social factors whatever parameter is under examination. Genetic analysis is inseparable from demographic and mating patterns; nutrition and energetics are inseparable from food production and food distribution, and land holding; climatic adaptation must encompass the technology of housing and clothing; biomedical fitness is related to population size, sanitary systems, health services and life-style (Weiner, 1982, p. 19).

Several other leading human biologists in the United Kingdom who were junior to Weiner and influenced by his ideas maintained this biocultural perspective. They are Geoffrey A. Harrison, Derek F. Roberts, and James M. Tanner. Roberts conducted pioneering demographic and genetic research on Southern Sudan peoples (Roberts, 1956). Harrison and Tanner, along with Barnicot and Weiner, wrote the first modern textbook in *Human Biology* in 1964, that continued the British tradition of anthropological and biocultural perspectives in human biology (Harrison et al., 1964). In each section of the book, culture figured prominently in examples that were presented. Overseas field experience and a commitment to comparative studies were attributes that exemplified most of this generation of biomedically trained human biologists. Each of these attributes led to an appreciation of the need to study human biology in the context of human behavior and culture.

In the United States, with the stimulus of Washburn's ideas on science and evolution, Earnest Hooton's former students at Harvard were moving the human biology of living populations forward. Washburn contributed very little to the growth of human biology because his interests were largely in skeletal biology, functional morphology, and primatology. Hooton's students, who completed their PhDs between 1940 and 1956, were to become the leaders in human biology during the next generation. These included, with PhD year in parenthesis: Alice Brues (1940) on the genetics of eye, skin, and hair; Marshall T. Newman (1941) on the peopling of the American Southeast; Joseph B. Birdsell (1942) on Australian Aborigine populations; Gabriel W. Lasker (1945) on Chinese migrants; James N. Spuhler (1946) on human genetics; Stanley M. Garn (1948) on human hair composition and distribution; William S. Laughlin (1949) on Aleut populations; Edward E. Hunt, Jr (1951) on Micronesian populations; and Paul T. Baker (1956) on desert-heat stress. With the exception of Alice Brues and Paul Baker, all of these young anthropologists attended some or most of the six Summer Seminars organized by Washburn that were held in New York City, and many contributed directly to these sessions, as well. Gabriel Lasker's (1999) memoir describes some of his associations with Hooton's students at this time.



3.5. Gabriel W. Lasker (1912–2002) at his desk. Photograph courtesy of Bernice A. Kaplan.

Again, in parallel with the United Kingdom, Americans Gabriel W. Lasker and Paul T. Baker provided inspiration and guidance to junior colleagues and students by promoting a biocultural framework. As noted, Lasker (Figure 3.5) had built on Boas's migration model in studies of Chinese (Lasker, 1946) and Mexican (Lasker, 1952) migrants. In addition to his research, which he pursued actively for more than 50 years, a major and long-term contribution to students and junior colleagues was by way of support of their papers submitted to the journal *Human Biology*, which he edited for 35 years. Baker (Figure 3.6), published widely with his own strong biocultural commitment, and also socialized his students about the value of biocultural research. In a paper published four and a half decades ago on "Climate, culture, and evolution," he stated: "To completely reject the concept of climatic selection without cultural involvement would be premature, but from anthropology's present theoretical framework it is more accurate always to consider the role of culture when trying to formulate an evolutionary process related to climate" (Baker, 1960, p. 5). Twenty years later

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3.6. Paul T. Baker (1927–2007) (left) and Stanley M. Garn (1922–2007) (right) some time during the late 1950s or early 1960s. Photograph courtesy of Barbara Garn.

Baker (1982) continued to emphasize this theme about the importance of the linkage between human biology and culture in a Huxley Memorial Lecture. In his 1996 Pearl Memorial Lecture, he summarized his ideas and underlined the "… need for all human biologists to have some training in relevant aspects of cultural anthropology" (Baker, 1997). Such a perspective seemed not only intuitively correct and logical to Baker, but it also led to more productive lines of research and testable hypotheses. Weiss and Chakraborty (1982, p. 383) observed that the complex influence of culture on evolution "… is a point still largely misunderstood or ignored by many researchers without anthropological training." Harrison (1982, p. 471) suggested that it is much easier to give "equal attention … to cultural as to biological …" processes in the United States than in Europe, because the subfields of anthropology can often be found in the same department in the United States.

Human biology research and student training during the immediate post-war years was supported enormously by the private Wenner-Gren Foundation for Anthropological Research (founded as The Viking Fund in 1941). Partly because of the close personal relationship between its Director, Paul Fejos, and Sherwood Washburn, the Foundation supported the Summer Seminars, the *Yearbook of Physical Anthropology*, and other activities associated with the development of physical anthropology during that time (Szathmáry, 1991). During these early years, and up to the present, it is hard to imagine how human biology, within the larger field of physical anthropology, could have developed into a mature science without the vision of Paul Fejos and later directors of the Wenner-Gren and the financial support of the Foundation (Baker and Eveleth, 1982).

AFTER THE MIDDLE OF THE TWENTIETH CENTURY

After 1950, new directions were taken in climatic morphology and physiology, genetics, disease, and adaptation to the environment. The book by Coon et al. (1950) outlined applications to climatic rules in humans including Bergmann's rule (body size) and Allen's rule (size and shape of extremities), and may have stimulated several biogeographic studies of human form. For example, Derek F. Roberts (1952, 1953) from the United Kingdom and Eugene Schreider (1950, 1951) from France explored relationships between body size and basal metabolism and climate in human populations around the world. They found that humans, as with other warm-blooded animals, conformed to Allen's and Bergmann's rules. Other anthropologists from both sides of the Atlantic conducted similar studies of the climatic environment and adaptation of humans to both heat and cold stress (Newman, 1953, 1956; Weiner, 1954; Newman and Munro, 1955; Baker and Daniels, 1956; Baker, 1958a, 1958b). Some of these studies were experimental (either laboratory or field) and were influenced by the physiological studies conducted of heat and cold stress of South Africans (Wyndham et al., 1952), Australian Aborigines (Scholander et al., 1958), Arctic Indians (Irving et al., 1960), Inuit/Eskimos (Brown and Page, 1952; Meehan, 1955), Alakaluf Indians (Hammel, 1960), and Laplanders (Scholander et al., 1957). These and other studies demonstrated quite clearly that human populations distributed around the world had adapted *morphologically*, *physiologically*, and *behaviorally* to the climatic conditions of their environments.

The difference between the physiologists' studies and the anthropologists' studies was in the anthropologists' understanding of the role of culture in these patterns of adaptation to temperature extremes (Baker, 1960). The anthropologists were more concerned with the whole culture/population and variations in responses by age and sex, whereas the physiologists' contributions were more focused on physiological mechanisms in small samples of men tested under controlled laboratory conditions. Both kinds of studies enriched what we learned and were complementary.

One of the post-war military institutions that stimulated work in anthropology and human biology was the Natick, Massachusetts Quartermaster Corps Climatic Research Laboratory (Francesconi et al., 1986). Being close to Harvard University it could draw on talented individuals from this university and also influence research directions there and elsewhere, as well. Many of the young scientists at the Climate Research Laboratory were environmental physiologists who became well-known scientists in later years, while a few of the scientists were trained in physical anthropology (e.g., Paul Baker and Russell Newman). Since the

Natick laboratories were also engaged in clothing and nutrition studies, there were also strong interests in body size (anthropometry) and body composition variations in military personnel. These government studies contributed to the mix of interests in nutrition, disease, body composition, and climate in the context of human adaptation to variable environments.

By the early 1960s, studies of body composition in human biology were on the rise (Brožek and Henschel, 1961; Brožek, 1963; Garn, 1963). Interests among anthropologists were derived, in part, from knowledge of skeletal biology and were linked closely to studies of nutritional adaptation (Brožek, 1956; Newman, 1960, 1962), and child growth (Garn and Shamir, 1958). Anthropometric (Garn, 1961) and physique (Brožek, 1956) surveys were carried out to evaluate nutritional status, and associations between growth and nutrition in stressful environments were studied (Roberts, 1960; Schraer and Newman, 1958). Josef Brožek (1999) provided a personal history of body composition studies in human biology.

During this period, human population genetics focused almost exclusively on blood polymorphisms. A good example was Alice Brues's (1954) important paper demonstrating that selection must have operated on the ABO blood group system because of the nonrandom distribution of ABO allele frequencies around the world. Electrophoresis was first used to separate proteins, and it was at this time that "... the staggering magnitude of [human] genetic variation ..." became apparent (Cavalli-Sforza et al., 1994, p. 3). One of the great discoveries of the 1950s was the relationship between the sickle cell gene and malaria and the protection afforded against malaria by the sickle cell heterozygote. James Neel (1915–2000) had worked out the genetics of sickle cell disease (Neel, 1949), and then described the anemias of sickle cell disease and thalassemia in his 15th Cold Spring Harbor Symposium paper (Neel, 1951). Neel proposed mutation or selection as hypotheses to explain the high prevalence of these dangerous anemias in the Mediterranean and in Africa, but it remained for Anthony Allison to show that the major force was natural selection. Allison (1954) demonstrated that the sickle cell gene was a balanced polymorphism maintained by the selective pressure of malaria. He described in some detail the history of his early work and the conditions surrounding the discovery in an autobiographical paper (Allison, 2002).

James Neel established the first human genetics department at the University of Michigan in 1956. At that time, this program was one of the strongest in the country, partly because of its connection with anthropology. William J. Schull worked with Neel in human genetics and had a joint appointment in anthropology, while James N. Spuhler (1917–1992) and Frank B. Livingstone (1928–2005) were conducting genetic research in the anthropology department. Another major contribution in human genetics in the 1950s was Livingstone's (1958) paper on the anthropology of sickle cell in West Africa. Based on fieldwork in Liberia and extensive literature research on West Africa, he demonstrated the relationships among malaria, sickle cell distribution, mosquito ecology, agriculture (forest transformation), and language and human migration. Only four years after Allison's (1954) publication, Livingstone had shown the importance of culture in structuring evolutionary change through its influence on malaria prevalence and gene-frequency distributions. Culture change had produced evolutionary change!

Other important genetics work in the 1950s and 1960s by anthropologists outside the United States and the United Kingdom included research by Jean Hiernaux (1966), from Paris, who studied blood-group genetics in African populations. Another major figure is Francisco Salzano (Salzano and Callegari-Jacques, 1988), from Brazil, who first did post-doctoral research with James Neel at the University of Michigan in the late 1950s, and who went on to become the major human geneticist to study tropical forest Native Americans in South America.

HUMAN ADAPTABILITY, ECOLOGY, AND INTERNATIONAL PROGRAMS

Among human biologists, the 1960s ushered in a mature sense of scientific problem solving, an increasing commitment to understanding evolutionary process and adaptation to the environment, and a growing interest in integrated, collaborative studies. In anthropology, broadly, there was a movement toward empiricism with interests in ecological approaches and cultural materialism. At the same time, human biologists were exploring ecological models in the context of adaptation to the environment (Baker, 1962; Little, 1982). In 1964, four distinguished British human biologists published the first edition of an important introductory textbook (Harrison et al., 1964). In this book, Geoffrey A. Harrison and Nigel A. Barnicot dealt with genetics and phenotypic variation, James M. Tanner covered human growth, and Joseph S. Weiner wrote the sections on human adaptation and human ecology. This was an important work because it defined the field of human biology, was synthetic, and it was state of the science for the 1960s. Weiner's final section on "Human Ecology" was the first definition of this important perspective in human biology: topics included nutrition, disease, climate, and population (demography), and a central emphasis of the work was on "ecological adaptive processes."

Two other important events took place in 1964. Firstly, the International Council of Scientific Unions (ICSU) (now the International Council for Science) in Paris established the International Biological

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Programme (IBP) with a planning phase from 1964 to 1967, a research phase from 1967 to 1972, and a synthesis phase to follow from 1972 to 1974. The orientation of this worldwide program was ecological, and its theme was "The Biological Basis of Productivity and Human Welfare." A number of sections of the IBP were designed to cover various components of ecology and ecosystems studies. A separate section called Human Adaptability (HA) was to cover "the ecology of mankind" from a variety of perspectives including health and welfare, environmental physiology, population genetics, child growth, anthropology, and demography (Weiner, 1965). Secondly, although some preliminary preparation had begun in 1962, an important HA planning conference was held in Austria at the Wenner-Gren Foundation Burg Wartenstein Conference Center in the July of that year. It was at that meeting that Weiner (1966) outlined the major categories of planned research, and the other, more than 20 internationally represented, participants reported on current knowledge over a wide variety of topics in human biology for North and South American, European, Asian, African, and Australian populations (Baker and Weiner, 1966). Weiner, who was International Convener (director) of the Human Adaptability Section of the IBP prepared a handbook (Weiner and Lourie, 1969) of standardized methods, and at the end of the IBP published a compendium of the completed research (Collins and Weiner, 1977). The planning and research that followed resulted in the participation of 40 nations, the completion of more than 230 projects, and several thousand publications under the HA banner. In a retrospective review of the HA research, there has been some criticism based on topical omissions, but the contributions far outweigh the shortcomings (Ulijaszek and Huss-Ashmore, 1997). Harrison (1997, p. 25) noted, in a contribution to the same review: "Notwithstanding its limitations, it played a major part in converting the old defunct physical anthropology into the vibrant and exciting component of biological anthropology as it is today." In addition, it is quite clear that the associations made between the British and American human biologists during the IBP were of remarkable value by serving to cross-fertilize ideas and to reinforce the biocultural and environmental perspectives shared by most of the participants (Baker, 1988).

One of the major conceptual contributions of the IBP was in the organization of multidisciplinary research (Little et al., 1997). Because complex ecosystems required broad expertise, many individuals from different fields of science were required to collaborate. Ecologists from the IBP were organized according to major ecosystems (e.g., tropical forests, arctic tundra, deserts, and grasslands), while some of the HA projects focused on single human populations in special environments. Two early projects were the Kalahari Research Group to study the !Kung Bushmen initiated in 1963 (not IBP affiliated; Lee and DeVore, 1976) and the Andean Biocultural Studies initiated in 1962 (IBP affiliated, Baker and Little, 1976). Two other important IBP projects included the International Studies of Circumpolar Peoples, a four-nation investigation of the health of Inuit/Eskimos in Alaska, Canada, and Greenland (Milan, 1980), Population Genetics of the American Indian, a project that focused on the Yanomama horticulturalists from Brazil and Venezuela (Neel et al., 1977), and the Solomon Islands Project, organized by Albert Damon and others at Harvard (Damon, 1974; Howells 1987). These and other HA projects were discussed in detail by Hanna et al. (1972) and from theoretical perspectives by Lasker (1969).

Other major contributions from the IBP were the surveys of human growth (Eveleth and Tanner, 1976) and human gene frequencies (Collins and Weiner, 1977) that were conducted on populations around the globe that contributed primary data. In many of these smaller Third World projects, research design and "problem orientation" were subordinated to data collection, but these important data sets, then, became available for comparative, historical, and in-depth analyses for future investigators (Weiner, 1977).

Following the close of the IBP in the 1970s, a new international program arose from UNESCO (United Nations Educational, Scientific, and Cultural Organization) called the Man and the Biosphere Program (MaB). Additional multidisciplinary projects were affiliated with the MaB program including the Multinational Andean Genetic and Health Project (Schull and Rothhammer, 1990) and the Samoan Migrant Project (Baker et al., 1986). Other projects, such as the South Turkana Ecosystem Project (Little and Leslie, 1999), the Ituri Forest Project (Bailey, 1991), and the Siberian Evenki Project (Crawford et al., 1992) followed somewhat later. These multidisciplinary projects promoted international collaboration, gathered fundamental data on populations that were on the brink of extinction, were instrumental in developing new field and experimental methods, and provided research training for a whole new generation of human biologists.

DNA ANALYSIS AND MOLECULAR GENETICS

Weiss and Chakraborty (1982) provided a detailed review of the history of human genetics up to the late 1970s. The late 1970s and 1980s, however, marked a dramatic transition in human population genetics from a reliance on phenotypic proteins and inference about genes (phenotype-based approach) to the direct measures of genes via DNA (the new molecular genetics). This transition was based on the new laboratory methods that enabled DNA to be extracted and purified from Blood and other tissues (Crawford, 2000). These

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methods included: use of restriction enzymes to clip nucleotide segments of DNA (restriction fragment length polymorphisms), DNA hybridization, and the polymerase chain reaction for amplification of DNA. Some of the new DNA methods were used in human biology for phylogenic reconstruction of our ancestors, to reconstruct human population movements over space and time, and for forensic and ancient DNA analysis. New methods of DNA extraction and analysis also led to the Human Genome Project and the controversial Human Genome Diversity Project.

Mitochondrial DNA (mtDNA), which is only transmitted through the maternal line and does not undergo recombination, has been very useful in a variety of approaches to human evolution (Cann, 1986). One of the most remarkable studies of mtDNA was the so-called "Mitochondrial Eve" observation that startled scientists interested in modern human origins. Cann et al. (1987) found in a sample of 147 people from around the world that "All these mitochondrial DNAs stem from one woman who is postulated to have lived about 200 000 years ago, probably in Africa" (Cann et al., 1987, p. 31). This and other DNA work led to the "Out of Africa" hypothesis on modern human origins (Stoneking and Cann, 1989; Vigilant et al., 1991). More recent research incorporated the Y chromosome, which is inherited through paternal lineages (Hammer, 1995).

A variety of studies have been conducted to trace population distributions and migrations in the historic and prehistoric past. Cavalli-Sforza et al. (1988) and Sokal (1988) combined genetic, archaeological, and linguistic information to reconstruct population expansion in Europe. This geographic genetic research continues up to the present (Barbujani, 2000). Another area of interest that had been dominated by archaeological evidence soon saw the application of genetic data to the question of the peopling of the New World (O'Rourke, 2000). Based on mtDNA and Y-chromosome markers, Merriwether (2002) suggested a single-wave migration, possibly originating from ancient Mongolia through Siberia and into the New World between 20 000 and 30 000 BP. All of these dates fall within the late Pleistocene when the Wisconsian glaciation covered parts of North America and the cold Beringia land bridge between Siberia and Alaska was exposed.

By the late 1980s, it was feasible to sequence the whole human genome of nucleotides and DNA on the human chromosomes. This culminated in the Human Genome Project (HGP), a massive DNA-sequencing effort that was estimated to cost several billion dollars. In the United States, the Department of Energy (DOE) and the National Institutes of Health (NIH) supported the HGP (Crawford, 2000). The work conducted by an international consortium began in 1990 and was completed in 2003, several years ahead of schedule. The principal concern of human population biologists about the HGP was that it would provide a *generic* genome; that is, little or no information would be provided on human genetic variation in individuals and populations around the world. Such variation was of value to studies of human evolution and population history, as well as applied research in the genetic bases for disease (resistance, susceptibility, environmental interactions, etc.) and genetic epidemiology.

In response to these concerns, the Human Genome Diversity Project (HGDP) was founded in 1991 under the leadership of L. L. Cavalli-Sforza, the distinguished Stanford University geneticist. Also, the project was identified as urgent because of the disappearance of many small Mendelian populations. The proposed HGDP sparked a controversy for a variety of factors: first, there was distrust by anthropologists of the geneticists in their conceptualization of race (linked to typological race and eugenics); second, there were concerns by Native American groups about their being exploited again, about race stereotyping, and about the potential use of DNA for commercial gain (patenting genes); third, there was an element of political naïveté of the organizing geneticists about conflicting beliefs about race and, initially, there were few anthropologists included in the program and its design (Reardon, 2005). Because of conflicts and protests the original plan to establish cell lines of DNA samples from 25 individuals from each of 400 populations was never satisfied, and funding was not forthcoming. Later the HGDP was revived under the title of HGDP-CEPH (Centre d'Etude du Polymophisme Humain), and now DNA is available for 1064 individuals from 52 populations. These DNA samples have begun to be used for research (Ramachandran, 2005).

During the past half century, there has been increasing scientific activity in what Derek Roberts (1965) first referred to as "anthropological genetics." Crawford (2007) outlined the differences between "anthropological genetics" and "human genetics," where the former incorporates a biocultural perspective, focuses on the population and, often, non-Western populations, and centers on questions of interest to anthropologists, particularly evolutionary questions. Research in molecular anthropology, genetic epidemiology, forensic anthropology via DNA analysis, human origins, and the history of human migration and dispersal are all areas of exploration that are being pursued by anthropological geneticists.

REPRODUCTION AND CHILD GROWTH

Interests in reproduction in anthropology date back to the early part of the twentieth century and are linked closely to two primary emphases in human biology:

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evolution/natural selection and human health. In evolution, differential fertility is one of the main driving forces of selection, and fertility (measured as number of live-born offspring) is also a prime indicator of adult health and well-being. Studies of growth are an extension of reproduction and, of course, date back in human biology to Franz Boas.

Pioneering studies by Sophie Aberle, who was trained in genetics at Stanford and medicine at Yale, were conducted of sex and reproduction in San Juan Pueblo Indians in New Mexico (Aberle, 1928, 1931). She was a member of the National Research Council Committee for Research in Problems of Sex (Aberle, 1953) and worked for the Bureau of Indian Affairs. From her research she found that the poor reproductive pattern of Pueblo Indians was not based on reduced fertility, which in fact was very high, but rather was based on extraordinarily high infant mortality rates (about 25%). Ashley Montagu (1905–1999) conducted early work on female sexual maturation by discovering post-menarcheal sterility in adolescent girls (Ashley-Montagu, 1939a, 1939b, 1946). He found that, for a year or more following menarche in girls, there was a markedly reduced fecundity or even sterility, and "... that puberty and the power to procreate are not synchronous events ..." (Ashley-Montagu, 1939b, p. 213). The sociologists, Kingsley Davis and Judith Blake (Davis and Blake 1956) formulated a groundbreaking model of factors affecting fertility in the context of social behavior. This provided the basis for systematic hypothesis testing and refinement of the model, which was done by Bongaarts (1978). Campbell and Wood (1988) further refined the model for "proximate determinants" of fertility in non-Western (noncontraceptive) as well as Western populations. One of the first studies to document an important variable, not considered by Davis and Blake (1956), was the central importance of breast-feeding in fertility control by Konner and Worthman (1980) in studies of !Kung Bushmen. This and other research work stimulated a number of anthropological studies of nursing, energetics, and fecundity in traditional populations (Bentley, 1985; Gray, 1994; Vitzthum, 1994).

An important series of studies was conducted in the 1960s and 1970s that explored relationships among energy balance (diet and activity), body composition (fat and muscle), and reproduction (age of menarche) in adolescent girls. The work by Rose Frisch (Frisch and Revelle, 1970; Frisch and McArthur, 1974) led to a hypothesis that there was a critical weight, and, then later, a critical level of body fat that both triggered menarche and maintained normal menstrual cycles. Frisch's ideas were severely criticized, partly because of her unyielding adherence to this basic hypothesis. Subsequent research has demonstrated that reproductive controls on the ovarian cycle are more complex than just being a function of the energy of body stores, although energy balance does play an important role. However, Frisch's research was truly pioneering in that it contributed to the development of a new *ecology of reproduction* that began to explore the evolution and ecology of reproductive function in a variety of Western and non-Western peoples (Howell, 1979; Ellison, 1990, 2001; Leslie et al., 1994). These and other investigations explored relationships among nutritional status and availability of food resources, disease, physical activity, endocrine function, body composition, behavior, growth, and reproduction – truly an integrated, behavioral and environmental science of human reproduction.

At the same time as this new research direction in fertility and reproduction was being taken, so were new discoveries being made in infant, child, and adolescent growth studies. Surveys of the worldwide variation in human growth, which were originally compiled from the IBP HA studies, synthesized the available knowledge on population variations in human growth up to the late 1980s (Eveleth and Tanner, 1976, 1990). Somewhat later, Michelle Lampl conducted new longitudinal research of individual growth patterns (Lampl et al., 1992) in an ingenious study of infant length, where some infants were measured every day for more than a year. She and her colleagues found that rather than being a continuous process as believed, growth proceeded in "incremental bursts" (saltatory growth) followed by periods of stasis. Some measurements of infants showed daily increases in length of up to 1 cm! This work was extended to adolescents, who also showed saltatory growth (Lampl and Johnson, 1993), and has led to new lines of research in bone and soft tissue growth and in the endocrine control of growth.

In the 1980s and 1990s, increasing research in growth has been directed toward health and the total life span from conception to senescence and death. This research has had both basic and applied components related to health across what has been called the "life span approach" to the study of human biology (Leidy, 1996). A general principle of this approach is that "... no single stage of a person's life (childhood, middle age, old age) can be understood apart from its antecedents and consequences" (Riley, 1979, p. 4). Activity levels, diet, and exposure to adverse substances during childhood all influence health during adulthood and senescence. It is also the case that longitudinal study of the same individuals through time, in the tradition of Franz Boas, is one of the best methods to demonstrate these life span correlations. David J. P. Barker (Barker et al., 1989; Barker, 1992) developed an extension of this approach to understanding the life span. He discovered from early twentieth-century archival birthweight data that the lower the birthweights of infants, the greater the risk of coronary heart disease in adulthood. He also found that lower birthweight increases the risk of hypertension,

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stroke, and adult diabetes. These findings led to what is now known as the "fetal origins hypothesis" of adult disease that is currently being explored in a variety of research lines today, particularly in the context of juvenile and adult obesity.

Another important developing research area in human growth focuses on the evolution of the unique properties of human growth (Bogin, 1997, 1999). Humans have a unique growth pattern that is similar to nonhuman primates, but differs to the extent that we have a long period of nursing and post-weaning dependence (birth to seven years of age), a period of rapid growth that defines adolescence, a delayed onset of reproduction, and menopause in females. These, and other unique attributes of human development, facilitate development of the brain, manual dexterity, and the social and sexual skills needed for life in a complex society. Evolution of growth has been studied in both living primate and paleoanthropology studies. Also within the evolutionary framework, "life history theory" centers on the allocation of energy to somatic functions, such as growth and maintenance of the body, and to reproductive functions, such as gestation, lactation, and child rearing (Hill, 1993). This research paradigm reintegrates areas of reproduction, demography, and growth in new and interesting ways.

BIOMEDICAL ANTHROPOLOGY, HEALTH, AND EPIDEMIOLOGY

There are deep traditions of medical studies in anthropology and human biology. As noted above, physical anthropology was linked to anatomy during the nineteenth and early twentieth centuries, and many physical anthropologists were employed in medical schools because of their expertise in gross anatomy. Some were also trained as physicians. These biomedical interests in physical anthropology expanded to include epidemiology, public health, child growth, and nutrition, all in the context of disease or other conditions of ill health. Johnston and Low (1984) identify biomedical anthropology as being (1) based on "... the application of anthropological theory to problems of health and disease" (1984, p. 215) (bioculturally centered); and (2) focused on a "biological outcome" (disease centered).

Many biomedical contributions have been made with anthropological knowledge in epidemiology, where population, the target disease, culture, and the environment intersect in complex ways. Frank Livingstone's (1958) research on sickle cell in West Africa was conducted from the anthropological side to the biomedical side, whereas Carleton Gajdusek's (1977) Nobel Prize winning research on Kuru in New Guinea was conducted from the biomedical side to be informed by anthropology. Correspondingly, Baruch Blumberg's (2006) Nobel Prize winning discovery of the Australia antigen and hepatitis B virus was carried out while investigators searched for serum protein polymorphisms in non-Western (anthropological) populations.

Human biologists have also taken advantage of unique opportunities to explore diseases or health threats in traditional, modernizing, and modern populations (Garruto et al., 1999). The stresses of high altitude hypoxia were explored in the Andes (Baker and Little, 1976; Schull and Rothhammer, 1990), and the high prevalence of amyotrophic lateral sclerosis and Parkinsonian dementia in the Chamorros population were studied in Guam (Garruto et al., 1989). Of increasing interest in human biology are the effects of modernization in non-Western populations, migration from traditional to modern societies, and life-style transitions in Western populations. For example, the nutritional and life-style transition in the West, particularly the United States, has led to high caloric intakes, reduced physical activity, and high rates of obesity, cardiovascular disease, hypertension, and diabetes. At the same time, the epidemiological transition has led to declines in infectious diseases and increases in chronic diseases of aging and the diseases just noted that are associated with life style. An exception to declines in infectious diseases is AIDS and other emerging diseases, some of which are a function of human population and social disruption (see Chapter 27 of this volume). All of these old and newly emerging diseases offer opportunities for human biologists to explore them in biological, cultural, and environmental frameworks.

SOCIETIES AND JOURNALS IN HUMAN BIOLOGY

There are several professional societies devoted to human biology. The oldest is the Society for the Study of Human Biology (SSHB), which was founded in Britain in 1958. About a decade later, the International Association of Human Biologists (IAHB) was founded at a Wenner-Gren Conference in Austria. The Human Biology Association (HBA) was established in 1974 (originally named the Human Biology Council), and most recently the American Association of Anthropological Genetics (AAAG) was founded in 1993. The affiliation of these societies with the three main journals in human biology is listed in Table 3.1. *Human Biology* is the oldest journal, dating back to 1929; the British *Annals of Human Biology* was founded in 1974; and the *American Journal of Human Biology* is the most recent publication, having begun in 1989. Among the three journals in human biology, there are about 2500 pages of articles and reviews published each year. Currently, *Human Biology* is the official publication of AAAG and publishes principally in population biology and

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TABLE 3.1. A chronology of physical anthropology and human biology journals, societies, and major editors.

1918	*American Journal of Physical Anthropology* founded and edited by Aleš Hrdlička until 1942
1929	*Human Biology* founded and edited by Raymond Pearl until his death in 1940
1930	American Association of Physical Anthropologists established
1946	*Yearbook of Physical Anthropology* founded by Sherwood L. Washburn, edited by Gabriel W. Lasker
1953	*Human Biology* edited by Gabriel W. Lasker until 1987
1958	Society for the Study of Human Biology (SSHB) established in the United Kingdom (UK)
1963	SSHB and journal *Human Biology* become affiliated with James M. Tanner co-editor in the UK
1967	International Association of Human Biologists established at Wenner-Gren Conference in Austria
1972	SSHB and *Human Biology* become disaffiliated because of disagreements with the publisher, Wayne State University Press
1974	*Annals of Human Biology* founded in the UK by the SSHB
1974	Human Biology Council (HBC) established and affiliated with *Human Biology*
1988	HBC and *Human Biology* become disaffiliated because of disagreements with the publisher, Wayne State University Press
1988	*Human Biology* edited by Michael H. Crawford
1989	*American Journal of Human Biology* founded and affiliated with the HBC
1990	*American Journal of Human Biology* edited by Robert M. Malina
1994	Human Biology Council name changed to Human Biology Association
1995	American Association of Anthropological Genetics becomes affiliated with *Human Biology*
2002	*American Journal of Human Biology* edited by Peter T. Ellison

genetics; *Annals of Human Biology* is the official publication of SSHB; and the *American Journal of Human Biology* is the official publication of HBA. Histories of each of these publications and their associations can be found in Tanner (1999), Crawford (2004), and Little and James (2005).

Three other societies and their journals are linked to human biology: the American Association of Physical Anthropologists (*American Journal of Physical Anthropology, AJPA*), the Society for the Study of Social Biology (*Social Biology, SB*), and The Galton Society in the United Kingdom (*Journal of Biosocial Science, JBS*) (Johnston and Little, 2000; Alfonso and Little, 2005). The *AJPA* publishes about 15–20% of its articles in human biology, and both the *SB* and *JBS* publish a majority of the papers on topics in public health and health-related demography. Over the years, the vast majority of editors of these six journals have been physical/biological anthropologists and human biologists.

WHAT IS HUMAN BIOLOGY?

From its earliest beginnings, human biology has made unique contributions to scientific inquiry for a number of reasons. Firstly, a principal pursuit has been to understand and explain the basis for human variation in all its dimensions, at the individual and the population levels. Secondly, because of ties with anthropology, explanations have drawn on knowledge of both the biology and the culture of humans. This has been a consistent practice dating back more than a hundred years to Franz Boas, the founder of American anthropology, and one of the principal founders of human biology. Thirdly, understanding of these biocultural relations has been couched in evolutionary and adaptational theory. That is, evolutionary theory contributes to or explains how humans interact with their environments. Fourthly, although genetics structures human variation, humans are flexible in their adaptation to the environment; hence, there is an acute awareness of human plasticity in response to the physical and the social environment. Finally, there has been a willingness among human biologists to incorporate knowledge and expertise from other fields within the biological, medical, and social sciences in order to enrich our understanding of this most complex of species – *Homo sapiens*.

DISCUSSION POINTS

1. Why is "race" an inappropriate concept to apply to an understanding of human variation?
2. Did Franz Boas's championing of plasticity and his studies of human migrants demonstrate that genetics played only a minor role in human variation?
3. Was all research in human biology halted during World War II? If not, then what kind of research was done?
4. Who were the principal figures in the United States who contributed to the modernization of human biology? And what were their major

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contributions? Who were the principal figures in the United Kingdom?

5. The 1960s were dominated by the International Biological Programme and its Human Adaptability Component that included a number of human biologists from around the world. What were some of the basic research themes and why was this an important vehicle for the promotion of human biology research?
6. An important transformation in human genetics studies occurred in the late 1970s. What was this transformation and how did it influence our interpretation of human variation?
7. Why is reproduction of interest to human biologists? What are some new areas of research?
8. Why is health of interest to human biologists? What can knowledge of human biology contribute to our understanding of HIV?
9. What are the journals that are devoted to human biology research? Do they differ in the kinds of articles that they publish?

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